

Rodrigo Morais

SENIOR DATA AND AI ENGINEER

Göttingen, Germany | Remote preferred | Hybrid nearby | No relocation
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Professional Summary

Senior Data and AI Engineer with 30 years in production software and more than seven years specializing in data platforms. Designs streaming, analytical, and AI systems with explicit quality, cost, and reliability controls. Recent results include 10x faster data processing, 99.8% usable market-data coverage, and 60% lower AI operating cost.

Technical Skills

AI Engineering	LangGraph, LangChain, typed classification, graph workflows, agents, model routing, guardrails, automated evaluations, observability
Data Engineering	Python, SQL, PySpark, Spark Structured Streaming, Kafka, Snowflake, dbt, PostgreSQL, Cassandra, CDC
ML and Statistics	pandas, NumPy, scikit-learn, time-series analysis, feature engineering, regression, KNN, K-means, Naive Bayes
Orchestration	Prefect, Airflow, recoverable pipelines, retries, backfills, data contracts, reconciliation
Platform and Software	Go, TypeScript, JavaScript, Clojure, Java, AWS, Docker, Kubernetes, Terraform, CI/CD

Professional Experience

Independent Software Data and AI Engineer

Germany | August 2025 to present

INVESTMENT MANAGER PERSONAL INVESTMENT MANAGEMENT PLATFORM

Built a private software platform for quantitative research, portfolio operations, market data, strategy evaluation, and controlled decision support.

- Reduced AI operating costs by 60% through domain-specific agents, graph workflows, capability-and-cost-based model routing, guardrails, reference datasets, automated evaluations, and production observability.
- Built typed decision and review workflows that combine data signals, graph execution, LLMs, evaluations, and explicit operational gates.
- Increased data-processing performance by 10x by replacing batch pipelines with PySpark streaming pipelines.
- Increased usable market-data coverage from 94% to 99.8% through multi-source ingestion, automatic fallback, validation, reconciliation, and recovery.
- Reduced portfolio reconciliation and reporting from 90 minutes to under 10 minutes through automated ingestion, reconciliation, calculations, and validation.

Data Engineer

Remote | February 2021 to July 2025

JUSTWATCH GMBH

Engineered production data systems for high-volume product, analytical, and operational workloads.

- Accelerated end-to-end data processing by 4x through automated ingestion, validation, testing, orchestration, and recovery.
- Moved business-critical refreshes from twice daily to near real time using PostgreSQL change data capture and Spark Structured Streaming into Snowflake.
- Reduced Snowflake costs by 34% by making workloads attributable, optimizing pipelines and SQL, and eliminating redundant processing.
- Reduced production data incidents by 72% through observability, infrastructure improvements, configuration hardening, and automatic recovery.
- Improved data quality by 30% by embedding validation, transformation, and data-contract checks directly into pipelines.

Professional Experience Continued

Quantitative Finance Developer

Remote | May 2020 to February 2021

TC TRADERS CLUB

Modernized market-data collection and analytical delivery for quantitative research.

- Reduced market-data latency from one day to 15 minutes and increased availability from 95% to 99.5% through validation, retries, monitoring, and recovery.
- Reduced data acquisition and preparation from hours to minutes by introducing Snowflake and dbt for analytical modeling and tests.
- Raised validated market-data correctness to nearly 100% by redesigning collection, cleaning, transformation, and testing workflows.

Senior Software Engineer Data Migration

Remote | January 2019 to May 2020

TRADESHIFT

Built distributed migration capabilities used by teams moving from a monolith to services.

- Developed a CQRS-based migration platform that let more than 10 teams adopt microservices while continuing to use existing PostgreSQL data safely.
- Reduced migration-related production incidents by 70% through idempotent processing, automated validation, replayable migrations, and standardized contracts using Clojure, RabbitMQ, and Docker.

Senior Full Stack Software Engineer

Berlin Germany | November 2016 to January 2019

BABEL

Developed cloud-backed product capabilities and modernized web application architecture.

- Reduced feature deployment time by 20% by automating AWS infrastructure and delivery with Terraform and CI/CD across API Gateway, Lambda, DynamoDB, S3, and RDS.
- Improved application performance by 40% through application and service modernization.

Earlier Engineering and Technical Leadership

Software Engineer Senior Engineer Technical Lead | ThoughtWorks, ITS4, Scopus IT, CWI, DBC and earlier companies | 1997 to 2016

- Led teams of up to six engineers and delivered projects for international clients including Gap, LATAM, Bradesco, and industrial organizations.
- Built data warehouses, reporting platforms, migration services, REST APIs, distributed applications, CI/CD pipelines, and modernization solutions across banking, ERP, industrial production, and web systems.

Education

M.Sc. Software Engineering | UNIBTA Instituto Brasileiro de Tecnologia Avançada | 2010

Service Oriented Architecture

B.Sc. Applied Mathematics | ULBRA Universidade Luterana do Brasil | 2008

Applied Mathematics for Computer Science

Languages

Portuguese native | English professional | Spanish intermediate | German A1